

Quantum Gravitation

Lecture 15

Final Project Presentations

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$m_g^{95} \sim 10^{-32} \text{eV}/c^2$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$checksum_{66}^{after} = pass$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$artifact = hash(pdf, env, log, table, data)$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Submission Preservation Standard; M32-r6 citation guide; Low-Energy Parameter Reference appendix P